

Chapter 5

Sequential Logic

opposed to combinational logic

5.1 Objectives

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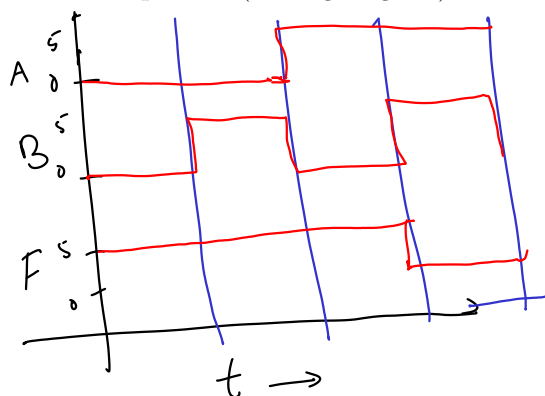
1. Understand timing diagrams, gate delays and critical path
2. Design Hazard-free two level circuits
3. Building blocks of sequential circuits
4. Analyze a sequential circuit and derive a state-table and a state-graph
5. Derive a state graph or state table from a word description of the problem
6. Understanding the structure of an FPGA

5.2 Why do we need sequential circuits?

Example 5.1. Think about this problem: Design an occupancy counter that depends on a sensor S at the class door. The sensor is triggered every time a person passes through the door. The counter can be reset to zero with a reset button. Assume we only need up to two bit counter C_1C_0 . Draw a truth table for this circuit. Do you have requisite knowledge for designing this circuit? Can this circuit be designed without a memory element?

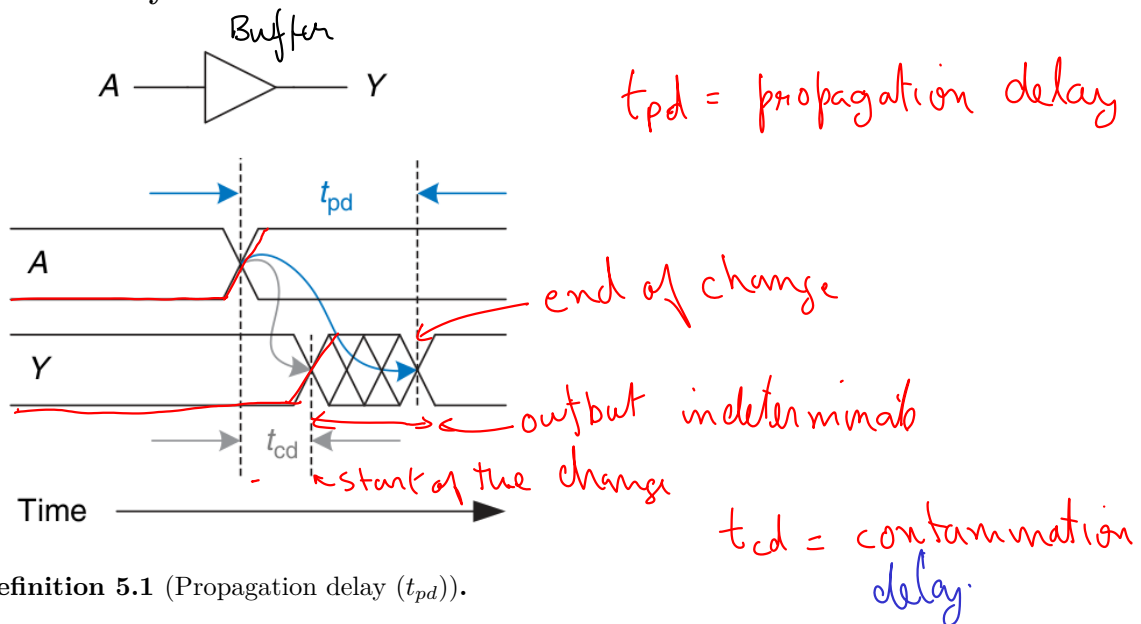
5.3 Timing diagrams and propagation delays

Example 5.2 (Timing diagram). Draw a timing diagram for an ideal NAND gate.



A	B	F
0	0	1
0	1	1
1	0	1
1	1	0

5.3.1 Delays



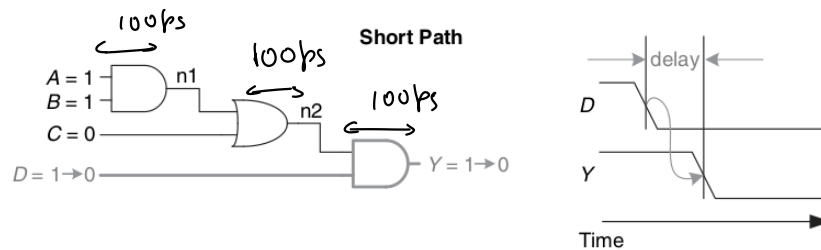
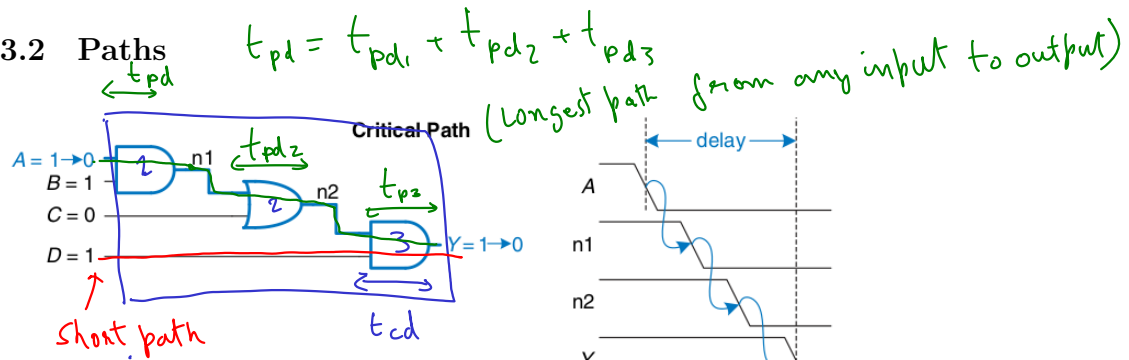
Definition 5.1 (Propagation delay (t_{pd})).

The minimum time it takes for the output signal to stabilize after the input changes.

Definition 5.2 (Contamination delay (t_{cd})).

The time it takes to modify the output signal after change in input

5.3.2 Paths



Example 5.3. Find the propagation delay of the circuit above given that propagation delay of each gate is 100ps and contamination delay of 60ps.

Example 5.4 (Ex 2.43 [1]). Determine the propagation delay and contamination delay of the circuit in Figure 5.1. Use the gate delays given in Table 5.2.

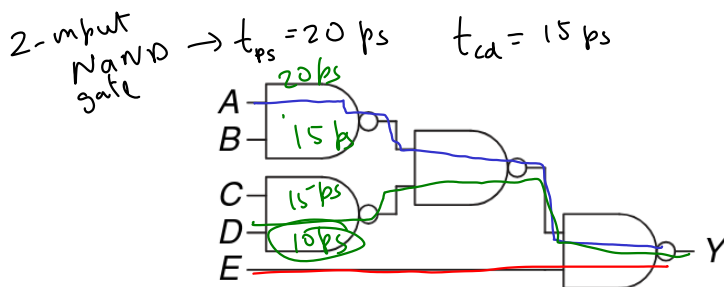


Figure 2.83 Circuit schematic

Circuit

$t_{ps} = 60\text{ps}$

$t_{cd} = 15\text{ps}$

Figure 5.1: Circuit

Table 2.8 Gate delays for Exercises 2.43–2.47

Gate	t_{pd} (ps)	t_{cd} (ps)
NOT	15	10
2-input NAND	20	15
3-input NAND	30	25
2-input NOR	30	25
3-input NOR	45	35
2-input AND	30	25
3-input AND	40	30
2-input OR	40	30
3-input OR	55	45
2-input XOR	60	40

Figure 5.2: Delays

Problem 5.1 (10 marks, Ex 2.44 [1]). Determine the propagation delay and contamination delay of the circuit in Figure 5.3. Use the gate delays given in Figure 5.2.

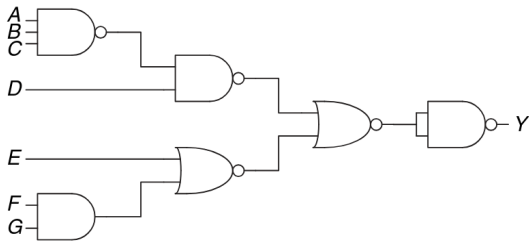
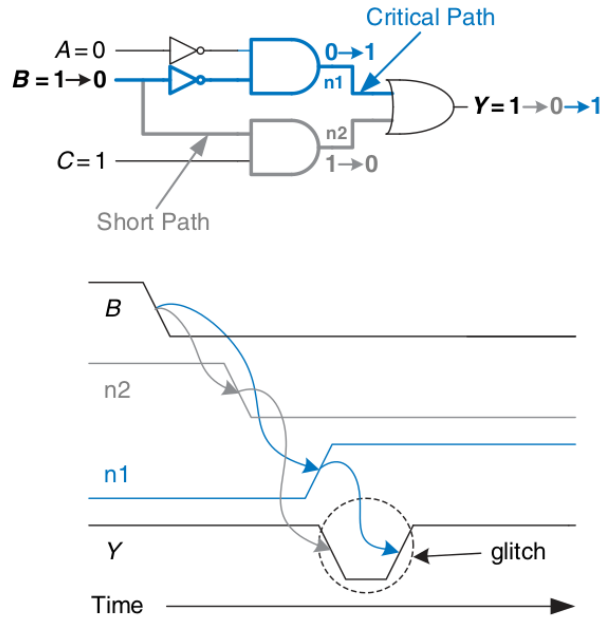


Figure 2.84 Circuit schematic

Figure 5.3: Circuit

5.4 Glitches or Hazards



Definition 5.3 (Glitch or Hazard).

Example 5.5. Design a circuit that fixes the glitch in the above circuit (also known as *glitch-free* or *hazard-free* circuit).

Problem 5.2. Find a minimal Boolean equation for the function in Figure 2.85. Remember to take advantage of the don't care entries (marked X) (10 marks).

A	B	C	D	Y
0	0	0	0	X
0	0	0	1	X
0	0	1	0	X
0	0	1	1	0
0	1	0	0	0
0	1	0	1	X
0	1	1	0	0
0	1	1	1	X
1	0	0	0	1
1	0	0	1	0
1	0	1	0	X
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	X
1	1	1	1	1

Figure 2.85 Truth table for Exercise 2.28

1. Sketch a circuit for the function (10 marks).
2. Does your circuit from have any potential glitches when one of the inputs changes? If not, explain why not. If so, show how to modify the circuit to eliminate the glitches (10 marks).

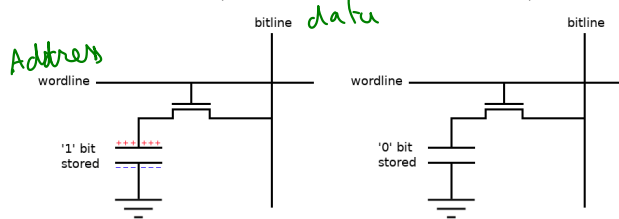
5.5 How to create memory element from circuits

Two types of memory

1. Volatile memory. For example, RAM, CPU registers.
2. Non-volatile memory. For example, SSD, Flash drives. (Not covered in this course)

volatile

- (a) Memories that require periodic refreshing. For example, DRAM: Dynamics Random Access memory (Not covered in this course)



1

- (b) Memories that are always refreshing. For example, SRAM: Static Random Access memory [3, Appendix B.64]

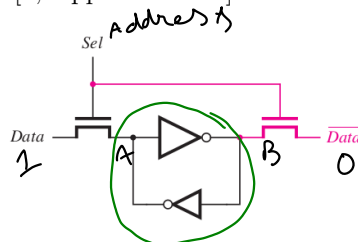
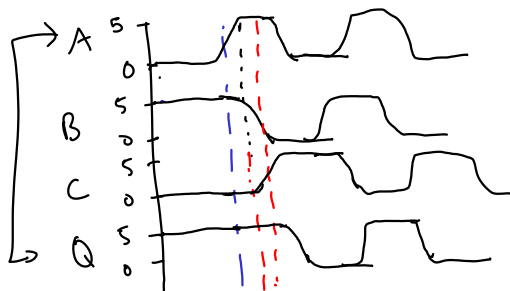
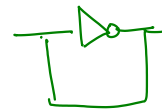
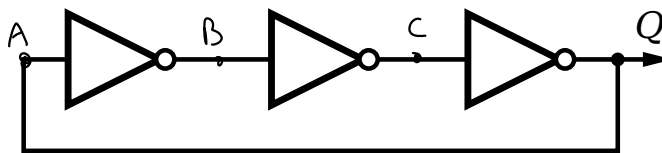


Figure B.64 An SRAM cell.

¹Image source: allaboutcircuits.com/technical-articles/introduction-to-dram-dynamic-random-access-memory/

5.6 Latches and Flip-Flops [1, Sec 3.2]

Example 5.6 (Ring oscillator). [1, Sec 3.31] How many stable states does the following circuit have?



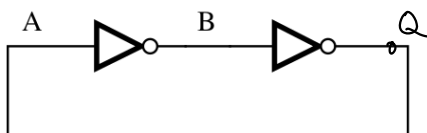
Ring oscillator

Astable circuit because
this has no stable
modes of operation
states

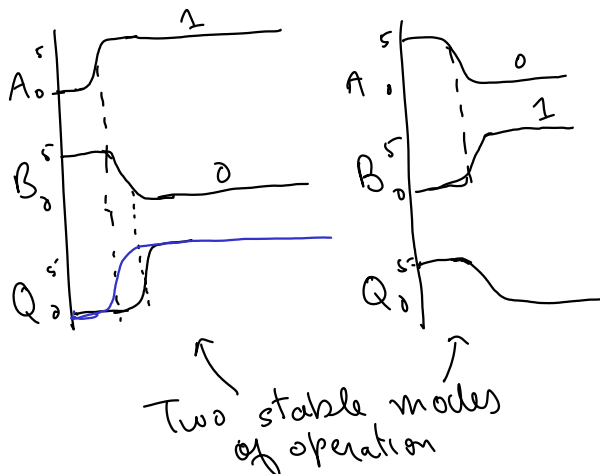
Definition 5.4 (Astable circuits).

Astable circuits have no stable "modes of operation" or states.

Example 5.7. Analyze the timing diagram of the following circuit.



Bistable circuit
2 states
1 bit of information

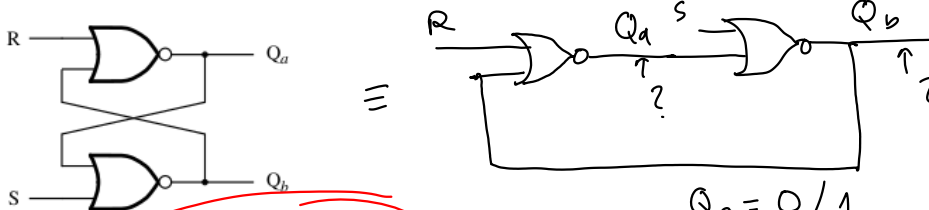


Definition 5.5 (Bistable circuits).

Definition 5.6 (Characteristic or state table). Draw the characteristic or state table of the above circuit.

5.6.1 SR (Set-Reset) latch [1, Sec 3.2.1]

Definition 5.7 (SR latch). The following circuit is called the SR latch.



Set = $S = 1$
Reset = $R = 0$

$$Q_a = 0/1$$

$$Q_b = \overline{S + Q_a} = 0$$

$$Q_a = \overline{R + Q_b} = \overline{0 + 0} = 1$$

Set = $S = 0$
Reset = $R = 1$

$$Q_a = \overline{R + Q_b} = \overline{1 + Q_b} = \overline{1} = 0$$

$$Q_b = \overline{S + Q_a} = \overline{0 + 0} = 1$$

- 2
1. How many stable states does this circuit have?
 2. Draw its characteristic or state table.
 3. Draw SR latch symbol

	S	R	Q_a^+	Q_b^+
Hold	0	0	Q_a	Q_b
Reset	0	1	0	1
Set	1	0	1	0
Invalid	1	1	0	0

$S = 0$
 $R = 0$

$$Q_a(t + \Delta t_{pd}) = \overline{R + Q_b(t)}$$

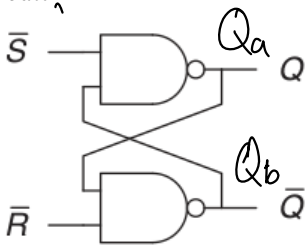
$$= \overline{0 + Q_b(t)} = \overline{Q_b(t)}$$

$$Q_b(t + \Delta t_{pd}) = \overline{S + Q_a(t)} = \overline{Q_a(t)}$$

$$Q_a = 1, Q_b = 0$$

$$Q_a = 1, Q_b = 0$$

Problem 5.3 (SR latch using NAND gates). Draw the characteristic or state table for the following circuit.



Case 1, 2, 3, 4
 $S = 0, R = 0$

$S = 0, R = 0$

$$Q_a(t + \Delta t_{pd}) = \overline{\overline{S} \cdot Q_b(t)}$$

$$= \overline{1 \cdot Q_b(t)}$$

$$= \overline{Q_b(t)}$$

$$Q_b(t + \Delta t_{pd}) = \overline{\overline{R} \cdot Q_a(t)}$$

S	R	Q^+	$\overline{Q}^+$
0	0		
0	1		
1	0		
1	1		

can contain
Q

$$S = 1$$

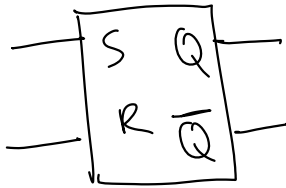
$$R = 1$$

$$Q_a(t + \Delta t_{pd}) = \overline{R + Q_b(t)}$$

$$= \overline{1 + Q_b(t)} = 0$$

$$Q_b(t + \Delta t_{pd}) = \overline{S + Q_a(t)}$$

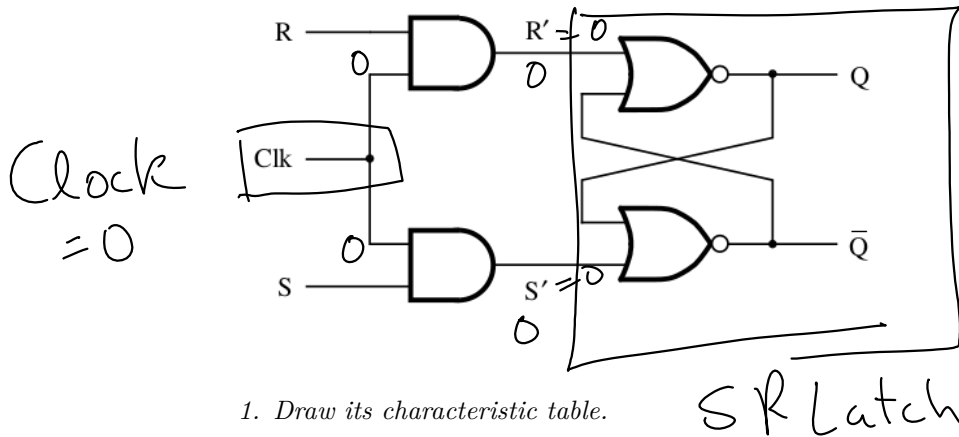
$$= \overline{1 + Q_a(t)} = 0$$



SR Latch

5.6.2 Gated SR latch [3, Sec 5.2]

Definition 5.8 (Gated SR latch). The following circuit is called the Gated SR latch.



1. Draw its characteristic table.
2. Draw the Gated SR latch symbol

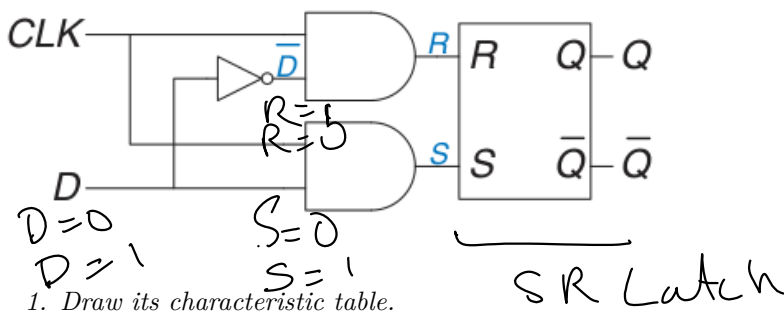
Clk	S	R	Q^+	
0	*	*	Q	Hold
1	0	0	Q	Hold
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	0	Invalid

Characteristic table

S'	R'	Q^+	
0	0	Q	Hold
0	1	0	Reset
1	0	1	Set
1	1	0	Invalid

5.6.3 D (Data) latch [1, Sec 3.2.2]

Definition 5.9 (D latch). The following circuit is called the D latch.



1. Draw its characteristic table.
2. Draw the D latch symbol

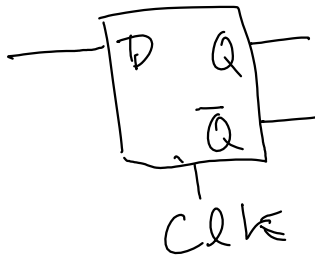
Clk	D	Q^+
0	*	Q
1	0	0
1	1	1

Transparent Latch

Clk	Q^+
0	Q
1	D

Memory

Hold state
Reset
Set



D-Latch Symbol

Problem 5.4. Consider the timing diagram in Figure 5.4. Assuming that the D and Clock inputs shown are applied to the circuit in Figure 5.5, draw waveforms for the Q_a , Q_b , and Q_c signals. (10 marks) [?, Prob 5.1]

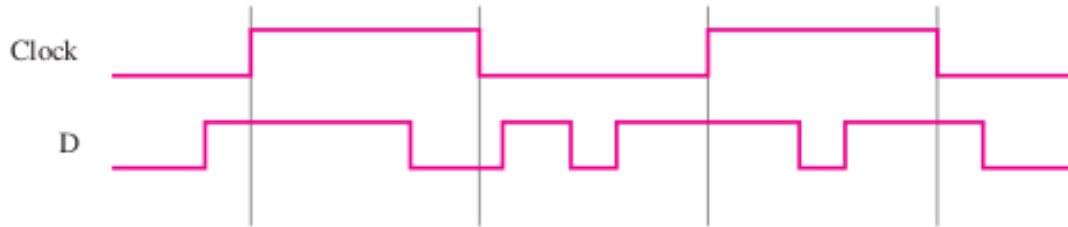
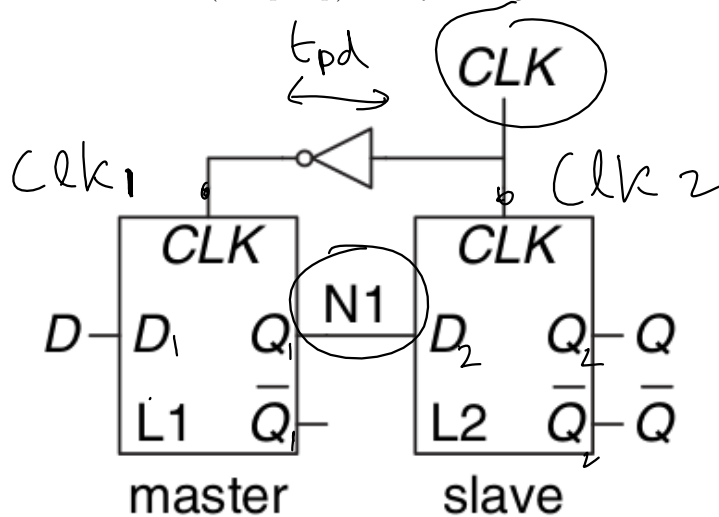


Figure 5.4: Inputs for Prob 5.4

5.6.4 D flip-flop [1, Sec 3.2.2]

Definition 5.10 (D flip-flop). The following circuit is called the D flip-flop.



1. Draw its timing diagram
2. Draw its characteristic table.
3. Draw the D flip-flop symbol

clock edges = pos edge
rising edge

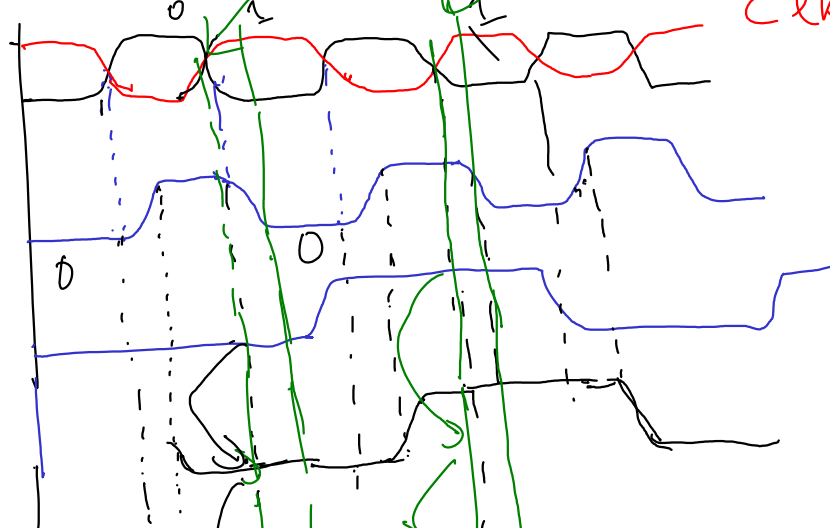
$clk2 = CLK$

$\overline{CLK2} = \overline{CLK}$

$CLK1$

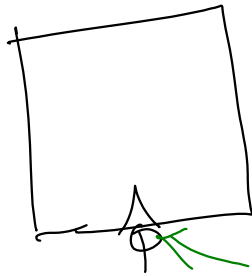
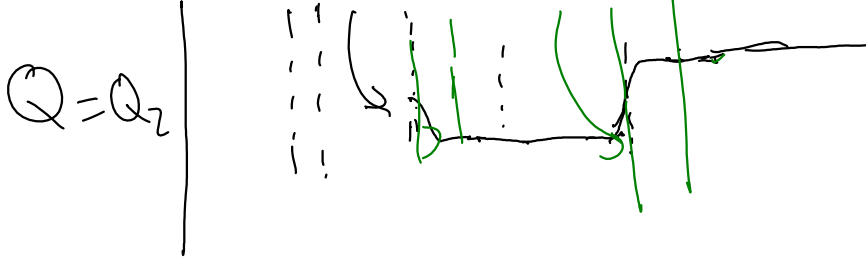
$D1$

$N1 = Q1$
 $= D2$

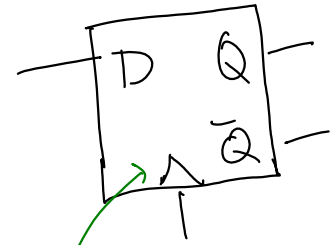


Characteristic table

$CLK \uparrow$	Q^+
0	Q
1	D



neg edge
triggered



edge CLK
triggered

FF = edge
triggered
Latch = level
triggered

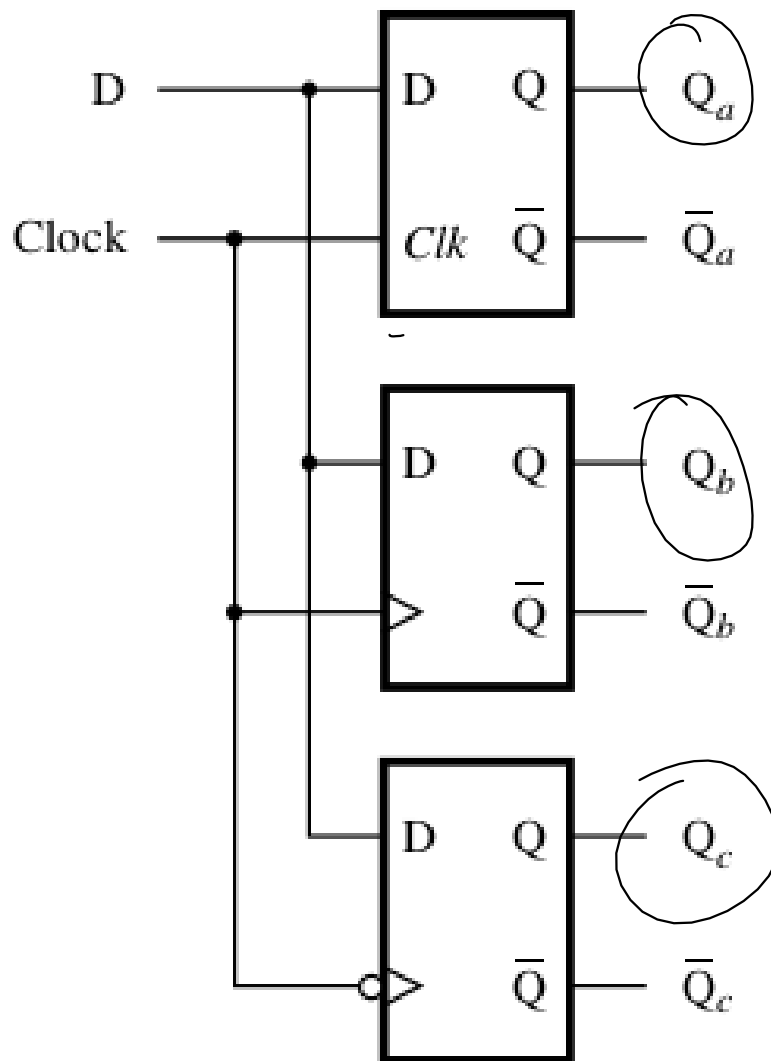
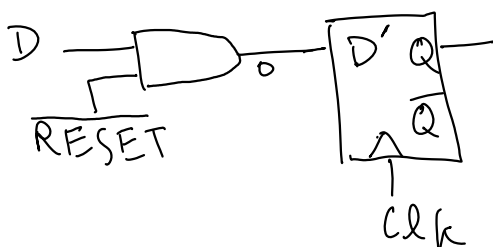


Figure 5.5: Circuit for Prob 5.4. Recall that Clk with a “ $\triangleright$ ” symbol indicates rising-edge (positive-edge) triggered flip-flop. Clk without “ $\triangleright$ ” symbol indicates a level-triggered latch. Clk with “ $\circ\triangleright$ ” symbol indicates a falling-edge (negative-edge) triggered flip-flop.

Remark 5.1. What is the difference between a latch and a flip-flop?

Example 5.8. Add a RESET signal to the D flip-flop that resets the state of flip-flop to 0.



Synchronous circuit are synchronised with clk (+ve edge of the clock)
Asynchronous Reset signal

Example 5.9. The toggle (T) flip-flop has one input, CLK, and one output, Q. On each rising edge of CLK, Q toggles to the complement of its previous value. Draw a schematic for a T flip-flop using a D flip-flop and an inverter.

3 kinds of ffs: D-ff, T-ff, JK ff

Data Toggle

Replace invalid with Toggle

D-ff		T-ff		JK ff		
clk	Q ⁺	T	Q ⁺	J	K	Q ⁺
0	Q	0	Q	0	0	Q
1	D	1	$\overline{Q}$	0	1	0
				1	0	1
				1	1	$\overline{Q}$

Hold Toggle

Problem 5.5. A JK flip-flop receives a clock and two inputs, J and K. On the rising edge of the clock, it updates the output, Q. If J and K are both 0, Q retains its old value. If only J is 1, Q becomes 1. If only K is 1, Q becomes 0. If both J and K are 1, Q becomes the opposite of its present state.

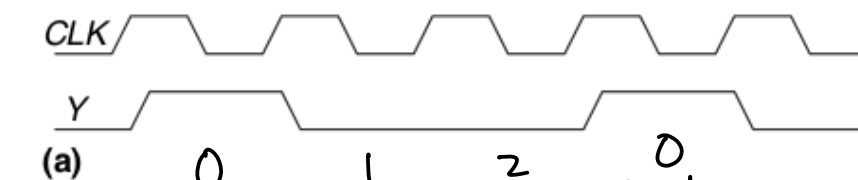
1. Construct a JK flip-flop using a D flip-flop and some combinational logic.
2. Construct a D flip-flop using a JK flip-flop and some combinational logic.
3. Construct a T flip-flop (see Exercise 3.9) using a JK flip-flop.

5.7 Finite State Machines [1, Sec 3.4]

2

Example 5.10. Design an occupancy counter that depends on a sensor S at the class door. The sensor is triggered every time a person passes through the door. Assume that the counter starts at zero. Assume we only need up to two bit counter C₁C₀. Draw a state table for this circuit.

Problem 5.6. A divide-by-N counter has one output and no inputs. The output Y is HIGH for one clock cycle out of every N. In other words, the output divides the frequency of the clock by N. The waveform for a divide-by-3 counter is shown here:



Sketch circuit designs for such a counter

²These notes will not fit on your note sheet.

$$2^1 < 3 < 2^2$$

State

Q₁ Q₀

Next State	Output
0 0	0
0 1	0
1 0	1
1 1	0

Q₁⁺ Q₀⁺

Next State	Output
0 0	0
0 1	0
1 0	1
1 1	0

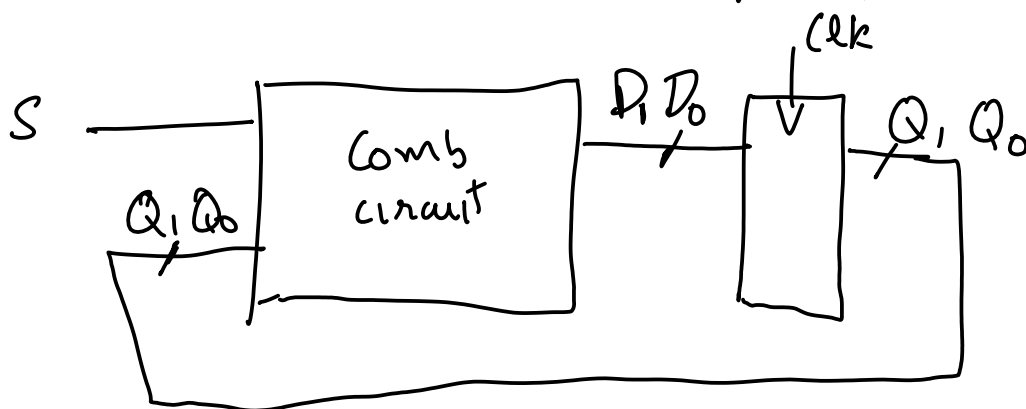
State transition table

Input S	State $Q_1 Q_0$		Next State $Q_1^+ Q_0^+$		D_1 $D_0 \leftarrow$ Truth table
	Q_1	Q_0	Q_1^+	Q_0^+	
0	0	0	0	0	
0	0	1	0	1	
0	1	0	1	0	
0	1	1	1	1	
1	0	0	0	1	
1	0	1	1	0	
1	1	0	1	1	
1	1	1	0	0	

D	Q	Q^+
0	0	0
0	1	0
1	0	1
1	1	1

Design a circuit (Use D-flip whenever not specified)

2-bit Register $\equiv$ 2-flip



$$D_1 D_0 = Q_1^+ Q_0^+$$

$$D(t) = f(Q(t), I(t))$$

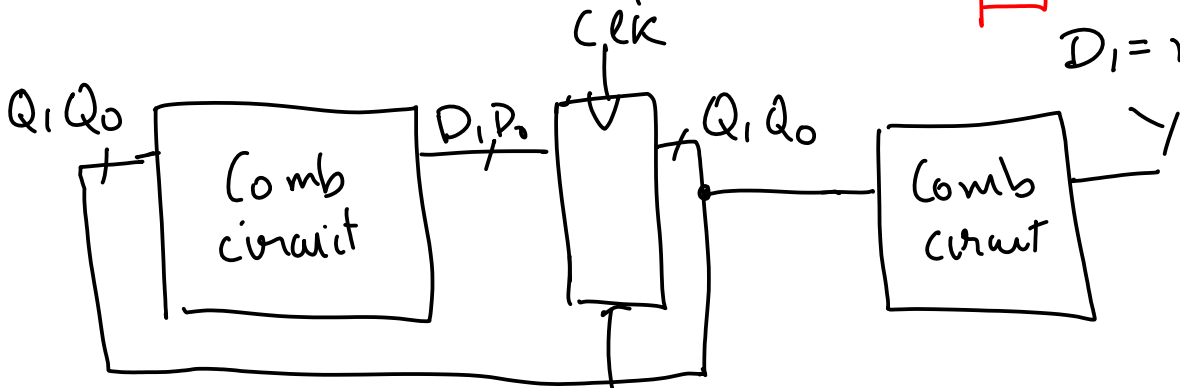
$$D_1 = Q_0 / D_0 = \bar{Q}_1 \bar{Q}_0$$

$$Q(t+1) = D(t)$$

State
 Q_1, Q_0
 0 0
 0 1
 1 0
 1 1

Next State	Output
Q_1^+, Q_0^+	Y
0 0	0
0 1	0
1 0	0
1 1	1

$$D_1 = m_1 = \bar{Q}_1 Q_0$$



D	Q	Q^+
0	0	0
0	1	0
1	0	1
1	1	1

2-bit D-register

$\equiv 2 \text{ D-ff}$

Q_0	Q_1
0	0
1	1

$$D_1 = Q_0$$

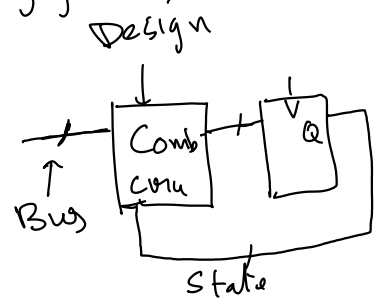
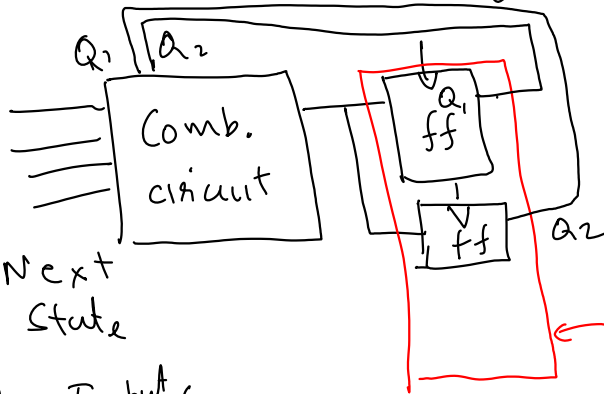
① Design of Sequential Circuits (Synchronous)

Design a T-ff using D-ff

Prob

Prev. State → Next State

Inputs + Prev State → Inputs to the Register/ff



A combination of ffs is called register

T-ff from D-ff

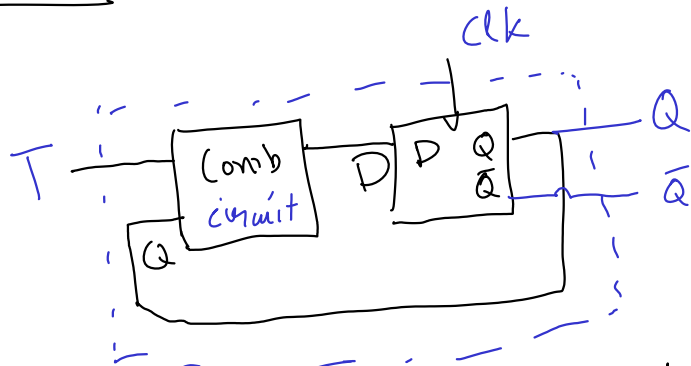
1 bit
2-states
1 ff

T	Q	Q ⁺
0	0	0
0	1	1
1	0	1
1	1	0

State transition table

Truth table

T	Q	D
0	0	0
0	1	1
1	0	0
1	1	1



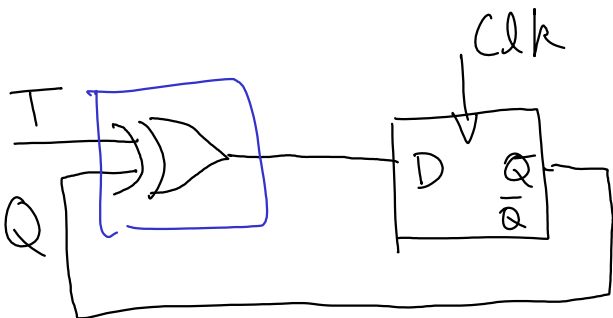
Q → Q ⁺	D
0 → 0	0
0 → 1	1
1 → 0	0
1 → 1	1

Excitation table

$$D = m_1 + m_2$$

$$= \bar{T}Q + T\bar{Q}$$

$$D = T \oplus Q$$



Practice : Design D-flip using T-flip

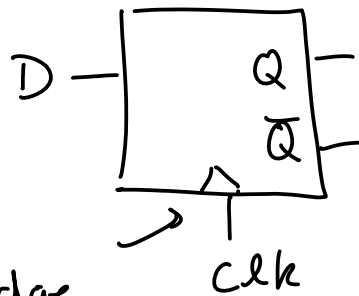
(1) Write State transition table for D-flip

(2) " Excitation table for T-flip

(3) Find Truth table for the comb circuit

Designing D-ff using J-K ff

D-ff



next state

edge triggered

characteristic table

D	Q^+
0	0
1	1

$Q^+ = D$

memorize

J-K ff

Set	Reset	Q^+
J	K	
0	0	Q
0	1	0
1	0	1
1	1	$\bar{Q}$

next state

hold

reset

set

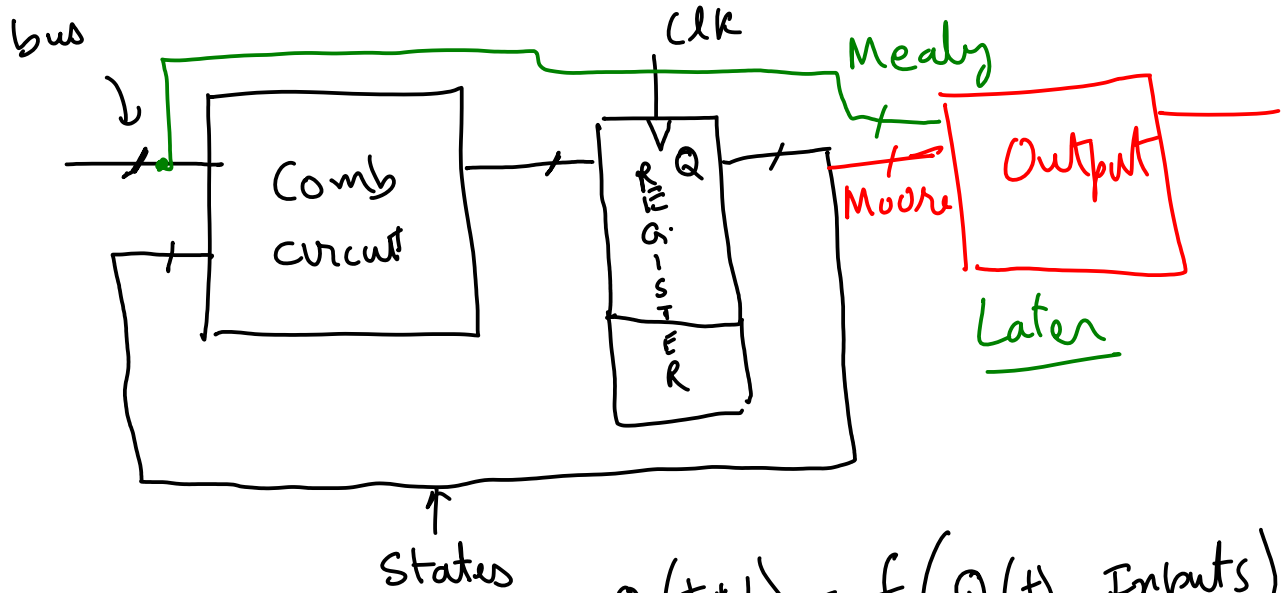
Toggle

Memorize

$$Q^+ = Q(t+1)$$

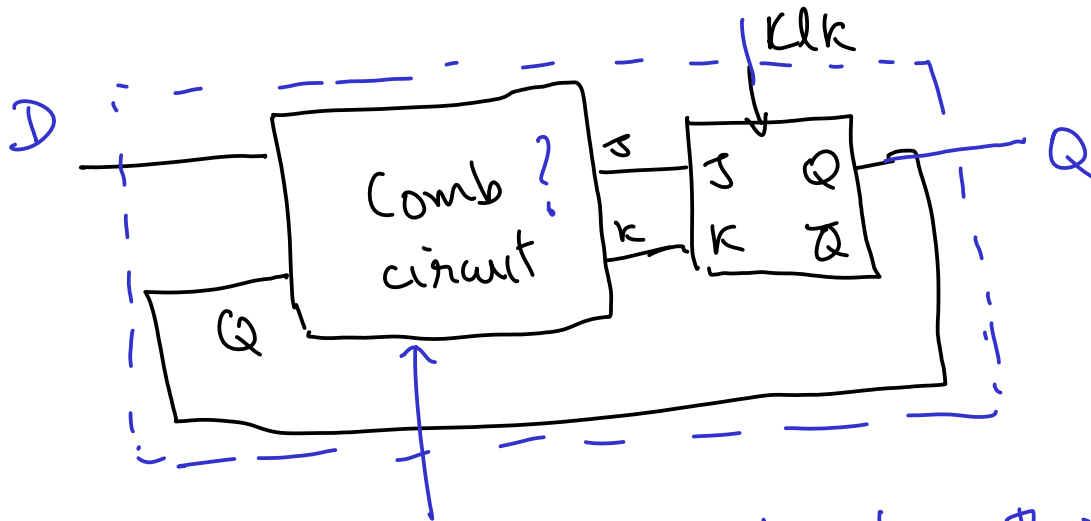
$$Q = Q(t)$$

Synchronous sequential circuit design



$$Q(t+1) = f(Q(t), \text{Inputs})$$

How many ff? D-ff has 1-bit memory
 J-K-ff has 1-bit memory
need 1-ff



find the truth table for this comb circuit

Truth table (find)

D	Q	J	K
0	0	0	0
0	1	0	1
1	0	1	0
1	1	1	1

State Transition table (desired)

D	Q	Q ⁺
0	0	0
0	1	0
1	0	1
1	1	1

FF excitation table

Q → Q ⁺	J	K
0 → 0	0	*
0 → 1	1	*
1 → 0	*	1
1 → 1	*	0

J	K	Q ⁺
0	0	Q
0	1	0
1	0	1
1	1	Q

0 → 0 hold
 0 → 1 reset
 1 → 0

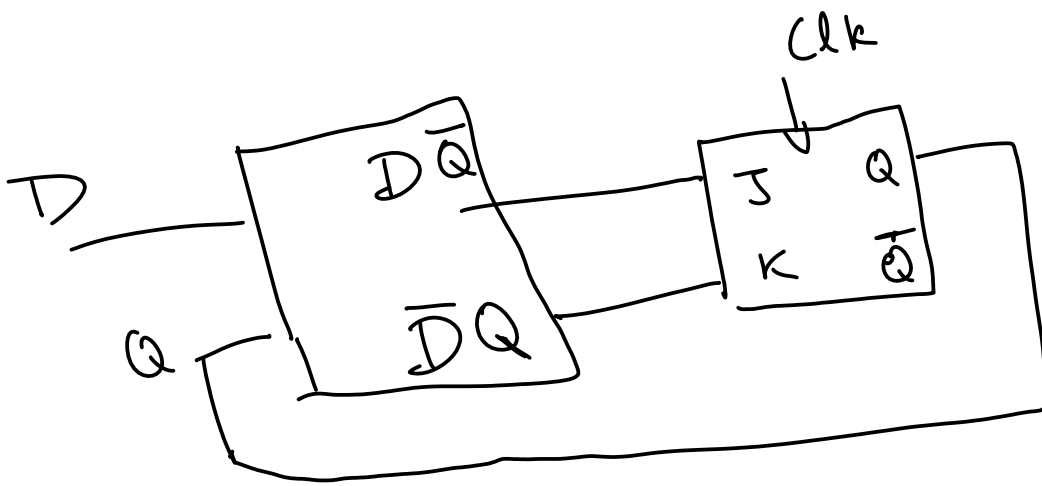
J	K
0	0
0	1
1	0
1	1

Q → Q⁺
 0 → 1 set
 1 → 1 toggle

what should JK values be for a particular Q-Q⁺ transition

$$J = D + Q = D\bar{Q}$$

$$K = \bar{D}Q$$



1. Draw the template circuit
2. Write the state transition table
3. Write the excitation table
4. Use the excitation table to convert state transition table to a truth table
5. Design the combinational circuit from the truth table

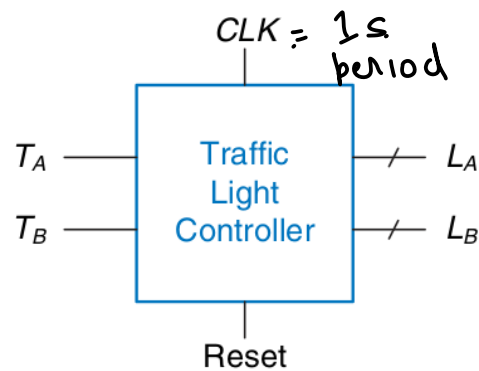
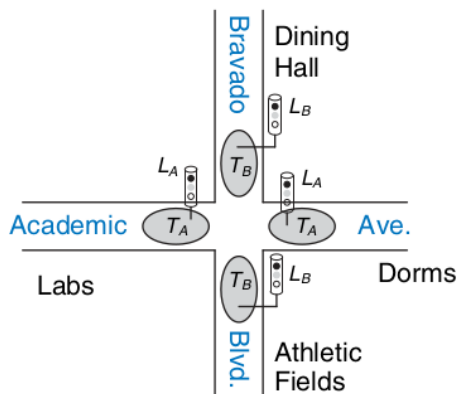
Problem 5.7. Design a 3-bit counter which counts in the sequence: 001, 011, 010, 110, 111, 100, (repeat) 001, ...

Example 5.11. Design an odd-even counter for an single bit input. The output of this circuit should be 1 if the number of 1s to the input have been odd so far and 0 otherwise.

Example 5.12 (Sequence detectors). A sequential circuit has one input and one output. The output becomes 1 and remain 1 thereafter when at least two 0's and at least two 1's have occurred as inputs regardless of the order of

Example 5.13. Consider the problem of inventing a controller for a traffic light at a busy intersection on campus. There are two traffic sensors, T_A and T_B , on Academic Ave. and Bravado Blvd., respectively. Each sensor indicates TRUE if students are present and FALSE if the street is empty. There are two traffic lights, L_A and L_B , to control traffic. Each light receives digital inputs specifying whether it should be green, yellow, or red. When the system is reset, the lights are green on Academic Ave. and red on Bravado Blvd. As long as traffic is present on Academic Ave., the lights do not change. When there is no longer traffic on Academic Ave., the light on Academic Ave. becomes yellow for 5 seconds before it turns red and Bravado Blvd.'s light turns green. Similarly, the Bravado Blvd. light remains green as long as traffic is present on the boulevard, then turns yellow and eventually red.

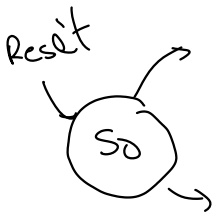
① State transition table/graph



1. Draw a state transition diagram
2. Draw a state table
3. Assign binary encodings to each of the states
4. Redraw the state table with binary encodings. Design a minimal SOP boolean expression.
5. Assign binary encodings to each of the output and redraw the output table. Design a minimal SOP boolean expression for the outputs.

Problem 5.8. Design a circuit for a 2×2 pixel resolution pong game, where the ball can only occupy 4 possible pixels and a single paddle occupies another 2 pixels. The ball bounces off the paddle when the paddle is in the correct row. To keep it interesting, the ball takes a different path from the source path. Track the score with a single bit counter.

State graph

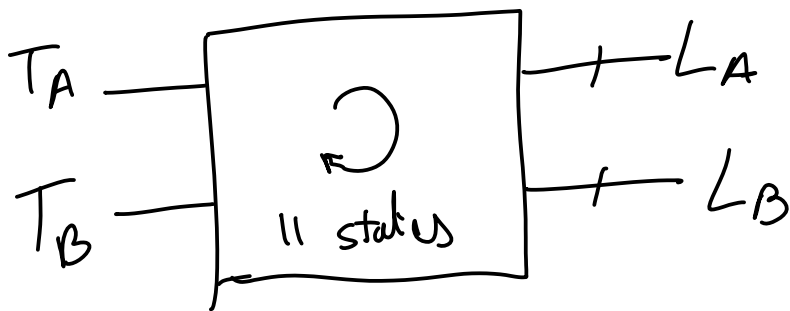


$$\begin{matrix} =00=01 \\ RYG=10 \end{matrix}$$

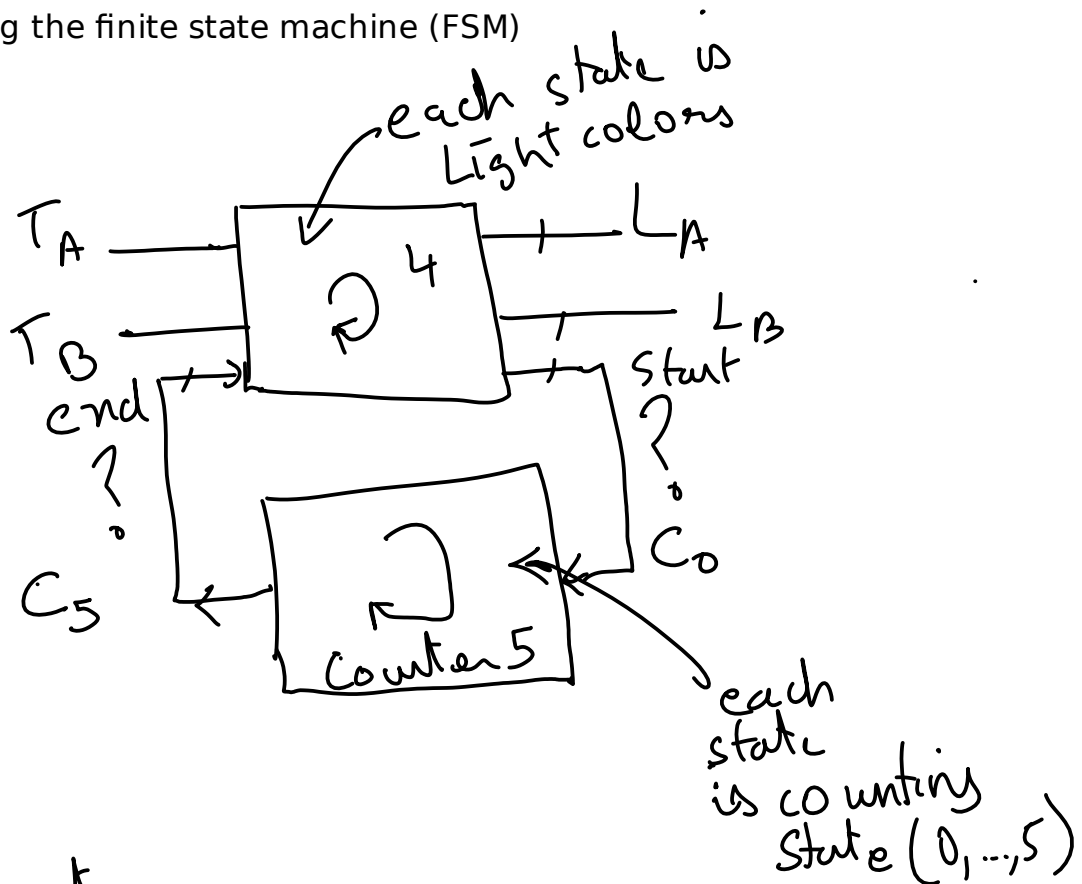
Red, Yellow, Green

Moore
Output
LA LB

State transition table		Next State				Moore Output LA LB	
State mean definition	Present State	Inputs TATB=00	01	10	11		
	S ₀	S ₁	S ₁	S ₀	S ₀	G	R
	S ₁	S ₂	S ₂	S ₂	S ₂	Y	R
0	S ₂	S ₃	S ₃	S ₃	S ₃	Y	R
1	S ₃	S ₄	S ₄	S ₄	S ₄	Y	R
2	S ₄	S ₅	S ₅	S ₅	S ₅	Y	R
3	S ₅	S ₆	S ₅	S ₆	S ₅	R	G
4	S ₆	S ₇	S ₇	S ₇	S ₇	R	Y
	S ₇	S ₈	S ₈	S ₈	S ₈	R	Y
	S ₈	S ₉	S ₉	S ₉	S ₉	R	Y
0	S ₉	S ₁₀	S ₁₀	S ₁₀	S ₁₀	R	Y
1	S ₁₀	S ₀	S ₀	S ₀	S ₀	R	Y
2							
3							
4							



Factorizing the finite state machine (FSM)



	Present State PS	Next State				Output		
		T_A	T_B	$= 00 \dots$		111	L_A	L_B
C_5								
*	S_0	S_1	S_1	S_0	S		G_R	0
*	S_1	S_2	S_2	S_2	S_2		Y_R	1
0	S_2	S_2	S_2	S_2	S_2		Y_R	0
1	S_5	S_4	S_5	S_6	S_5		R_G	0

State transition table $2^2 \leq 6$ states $\leq 2^3$ bits

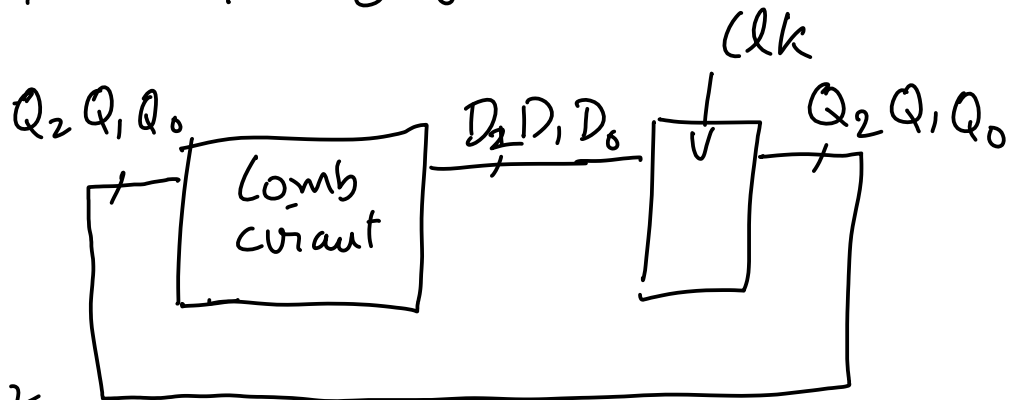
State				Next State		
Q_2	Q_1	Q_0		Q_2^+	Q_1^+	Q_0^+
0	0	0		d	d	d
0	0	1		0	1	1
2	0	1	0	1	1	0
0	1	1		0	1	0
1	0	0		0	0	1
1	0	1		d	d	d
1	1	0		1	1	1
7	1	1	1	1	0	0

Truth Table

Q_2, Q_1, Q_0

$$D_2 = m_2 + m_6 + m_7$$

$$D = Q^+$$



D_2	d	1	1	0
Q_0	0	0	1	d
	Q_1			

$$D_2 = Q_1 \bar{Q}_0 + Q_2 Q_1$$

Example 1. A sequential circuit has one input (X) and one output (Z). The circuit examines groups of four consecutive inputs and produces an output $Z=1$ if the input sequence 0010 or 0001 occurs. The sequences can overlap. Draw both Mealy and Moore timing diagrams. Find the Mealy and Moore state graph.

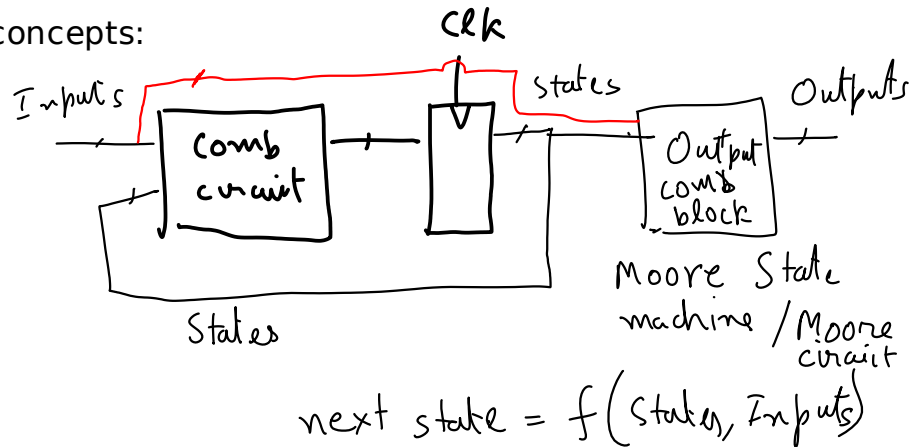
New terminology and concepts:

Mealy:

Moore:

State graph:

State encoding:



Moore state machine/Moore circuit: when output depends only on the states.

Mealy state machine/Mealy circuit: when output depends both on the states and the inputs.

Mealy outputs will change instantaneously when the inputs change.

Moore has the advantage of being synchronized with the clock. For Moore the outputs change only at the rising edge of next clock cycle.

Moore is slower; needs more memory/states/flip flops.

Additional level of complexity:
State encoding:

binary representation of the State

bits for the state
 $Q_2 Q_1 Q_0$

0	1	1
1	0	1
1	1	0

Design choice
Output

Design choice $\rightarrow$ $\begin{matrix} Q_2 & Q_1 & Q_0 \\ \hline 0 & 0 & 0 \end{matrix}$ $\rightarrow$ $\begin{matrix} Z_2 & Z_1 & Z_0 \\ 0 & 1 & 1 \end{matrix}$

state encoding can be different from output

MOORE circuit (Finite State Machines)

(Finite State Machine + Tape Turing Machine)

Detect 0010 or 0001 with overlap

clk
 x 01011100100100 ← Example sequence
 z 0000 0 00000100101 ← Desired output

In a Moore circuit the output changes at the next clock cycle after the input changes.

Do not solve for State encoding in the first Step

Use alphabets to denote unique States $S_0, S_1, S_2, \dots$
 $A, B, C, D, \dots$

State transition table

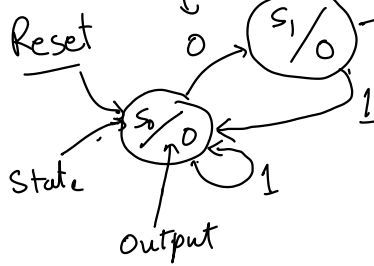
Input, State → Next State, Output

Meaning
 Define
 state

$S_0 = ?$
 $S_1 = ?$

(moore) State graph

(Mealy) Inputs (on arrows)



00010
 0001

0010 or 0001

00100
 00101

***1
 **01

Define

$S_0 = ****$
 $S_1 = ***0$
 $S_2 = ***1$
 $S_3 = **00$
 $S_4 = *001$
 $S_5 = 0010$
 $S_6 = *000$
 $S_7 = 0001$

State graph $\equiv$ State transition table
 +
state encoding
 =
State encoded trans. table

State encoded → optimally assign state encoding
 k-map based strategies

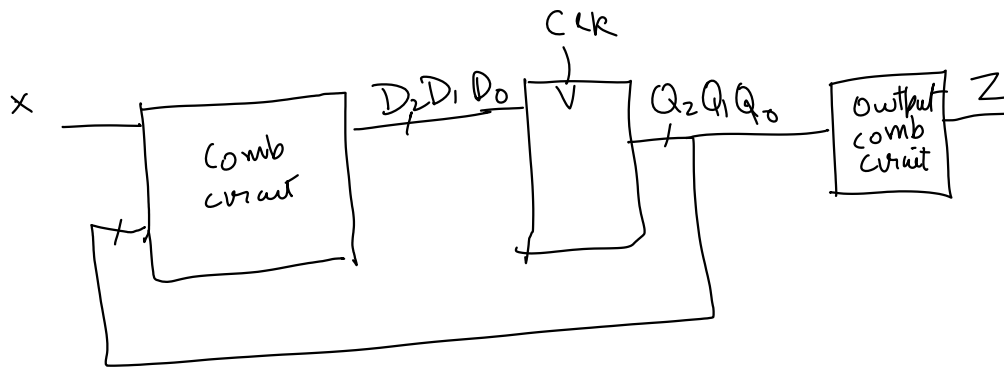
For today just assign any state encoding

of states = 7 → # of bits / # of ff

$2^2 = 4 < 7 \leq 8 = 2^3$
 3-bits

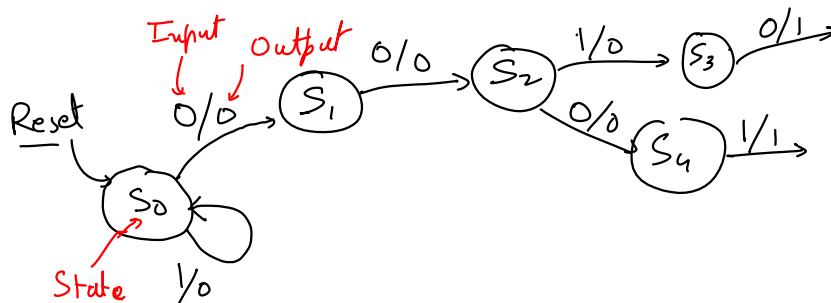
State	encoding	Input	State encoded	trans table	output
	$Q_2 Q_1 Q_0$	X	$Q_2 Q_1 Q_0$	$Q_2^+ Q_1^+ Q_0^+$	Z
S_0	0 0 0	0	0 0 0	0 0 1	0
S_1	0 0 1	0	0 0 1		
S_3	0 1 0	0	0 1 0		
S_4	0 1 1	0	1 0 0		
S_5	1 0 0	0	1 0 1		
S_6	1 0 1	0	1 1 0		
S_7	1 1 0	0	1 1 1		
		1	0 0 0		
		1	0 0 1		
		1	0 1 0		
		1	0 1 1		
		1	1 0 0		
		1	1 0 1		
		1	1 1 0		
		1	1 1 1		

D-flip



1. Define states
2. Draw state graph
3. Assign state encoding
4. Write state encoded trans. table
5. Convert it to Truth table using excitation table
6. Design the required combinational circuits

Mealy State graph Detect 0010 with overlap



Define
 $S_0 = ****$
 $S_1 = * * * 0$
 $S_2 = * * 0 0$
 $S_3 = * 0 0 1$
 $S_4 = 0 0 1 0$
 $S_4 = * 0 0 0$

Example 5.12 (Sequence detectors). A sequential circuit has one input and one output. The output becomes 1 and remains 1 thereafter when at least two 0's and at least two 1's have occurred as inputs regardless of the uorder. count count .